

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I _____B.Sreenitha_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B.Sreenitha

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $x: \text{Vehicle}(x) \text{ EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall \forall x: \forall \text{ClearPath}(x) \wedge \rightarrow \wedge \text{GreenSignal}(x) \wedge \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $x y: \text{Vehicle}(x) \text{ Behind}(x,y) \text{ GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \text{ CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let the propositions for **Patient A** be:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **R** = Patient A has Rash
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Possible Flu
- **M** = Patient A has Possible Measles
- **P** = Patient A has Possible Pneumonia

Rules

Rule 1: Fever AND Cough $\rightarrow$ Possible Flu

$$F \wedge C \rightarrow \text{Flu}$$

Rule 2: Fever AND Rash $\rightarrow$ Possible Measles

$$F \wedge R \rightarrow M$$

Rule 3: Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

$$C \wedge B \rightarrow P$$

Facts

F=True

C=True

B= True

There is **no fact that Rash is true**, so:

R=False/Unknown

Symbols

Symbol Meaning

$\wedge$ AND

$\vee$ OR

$\rightarrow$ Implication / IF–THEN

$\neg$ NOT

F,C,R,B Patient symptoms

Flu,M,P Possible diagnoses

(b) Forward Chaining

Forward chaining starts with known facts and repeatedly applies rules whose conditions are satisfied.

Initial facts

F,C,B

Step 1 — Apply Rule 1

Rule:

$F \wedge C \rightarrow \text{Flu}$

We know:

$F = \text{True}, C = \text{True}$

Therefore:

Derived fact: Patient A may have Flu.

Step 2 — Apply Rule 2

Rule:

$F \wedge R \rightarrow M$

We know:

But we **do not know** $R = \text{True}$.

Therefore, Rule 2 **cannot fire**.

M cannot be derived

Step 3 — Apply Rule 3

Rule:

$C \wedge B \rightarrow P$

We know:

$C = \text{True}, B = \text{True}$

Therefore:

$P = \text{True}$

Derived fact: Patient A may have Pneumonia.

Final Forward-Chaining Result

Measles cannot be derived because Rash is not known to be true.

Inference table

Step	Rule	Conditions	New Fact
0	Facts		Initial facts
1	Rule 1		
2	Rule 2		Not fired
3	Rule 3		

(c) Backward Chaining for Pneumonia

Goal

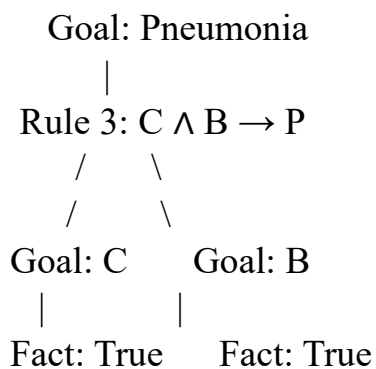
Backward chaining starts from the goal and works **backwards** to find the facts required to prove it.

The rule that concludes Pneumonia is:

Therefore, to prove Pneumonia, we need:

AND

Goal Reduction Tree



Therefore:

and

Both subgoals are satisfied.

Hence:

Conclusion

The goal is successfully proved using **Backward Chaining**.

Case Study 2 — Autonomous Traffic Management Agent

We have the following FOL knowledge base.

Rule 1

Rule 2

Rule 3

Facts

(a) Unification of Rule 1

Rule 1:

We compare the rule premises with the facts.

First predicate

Given:

Therefore:

So the substitution is:

Second predicate

Rule requires:

After substitution:

This fact exists.

Therefore:

This is the **Most General Unifier (MGU)**.

Rule 1 after substitution

Since both facts are true:

is derived.

(b) Resolution Proof of Proceed(CarA)

We want to prove:

Using **Resolution Refutation**, we assume the opposite:

and show that this leads to the empty clause:

Step 1 — Convert Rule 1 to CNF

Original:

Using:

we get:

Step 2 — Convert Rule 2 to CNF

Original:

This gives two clauses:

and

Step 3 — Convert Rule 3 to CNF

Original:

Therefore:

Step 4 — Ground the relevant clauses

From Rule 1 and the ambulance facts:

Resolve:

with the two facts.

We obtain:

Step 5 — Derive GreenSignal(Ambulance)

Rule 2:

We have:

Resolution gives:

Step 6 — Use Rule 3

Rule 3 with:

and

becomes:

We have the facts:

Resolving these successively gives:

Resolution Refutation

Add the negated goal:

But from the KB we derived:

Therefore:

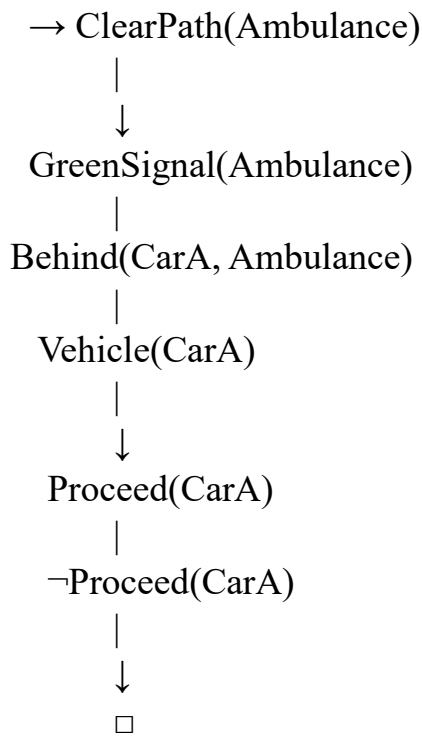
and

resolve to:

Hence:

Resolution proof tree

Vehicle(Ambulance) EmergencyType(Ambulance)
 \ /
 \ /



Thus, **CarA can proceed** according to the given knowledge base.

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle **multiple emergency vehicles individually**, because Rule 1 is universally quantified:

For example, if there are:

and

Rule 1 can derive:

and

However, the KB does **not explicitly define how to prioritize two emergency vehicles**.

Additional rules are needed

For example:

A collision/conflict rule could also be added:

We may also need facts such as:

Conclusion

The existing KB can recognize both emergency vehicles, but it is **not sufficient for conflict resolution or priority management**.

Therefore, additional **priority, conflict-resolution, and routing rules** are required.

Case Study 3 — Agricultural Expert Reasoning System

Given:

P1

P2

P3

Facts

(a) Convert to CNF

P1

Original:

Eliminate implication:

Therefore:

This is already CNF.

P2

Original:

Eliminate implication:

Using De Morgan's law:

Therefore:

This is CNF.

P3

Original:

Eliminate implication:

Using De Morgan's law:

Therefore:

Facts in CNF

The facts are already single-literal clauses:

Complete CNF KB

(b) Resolution Refutation for ApplyDripMethod

Goal

For resolution refutation, negate the goal:
Add it to the KB.

Step 1

From:
and
Resolve:

Step 2

Use:
and:
Resolve:

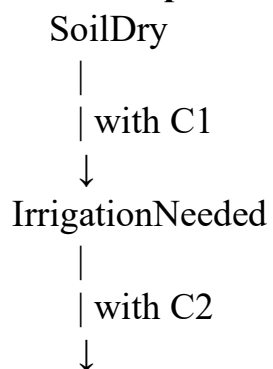
Step 3

Use:
with:
Resolution gives:

Step 4 — Contradiction with negated goal

We added:
Now resolve:
with:
Therefore:
The empty clause proves the theorem.

Resolution proof tree



$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

|

| with CropWheat

↓

ApplyDripMethod

|

| with $\neg \text{ApplyDripMethod}$

↓

□

Therefore:

(c) Remove SoilDry

Now remove the fact:

from the KB.

The remaining facts are:

The rule:

still exists, but we no longer know that SoilDry is true.

Therefore, we **cannot derive**:

Without IrrigationNeeded, Rule P2 cannot fire:

Therefore:

Does CropAtRisk follow?

Rule P3 is:

To derive CropAtRisk, we need:

and

We still know:

But we **do not have**:

The absence of a proof for ApplyDripMethod does **not** automatically mean that ApplyDripMethod is false.

This is an important logical principle:

Therefore:

Final comparison

Situation	Conclusion
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SoilDry present	IrrigationNeeded derived
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Situation	Conclusion
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SoilDry present	ApplyDripMethod derived
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SoilDry removed	IrrigationNeeded not derived
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SoilDry removed	ApplyDripMethod not derived
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SoilDry removed	not given
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SoilDry removed	CropAtRisk not derived
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Important point

We cannot say:

merely because ApplyDripMethod cannot be proven.

So: $\neg \text{ApplyDripMethod}$

CropAtRisk does not follow from the modified KB